Fatemeh Fahimi - 250554171








Site Analysis

Scale: 1/5000

- Industrial Building
- Residential Building
- Construction Area
- Electrical Substation
- Flooding Zone 1
- Commercial Building
- Green Area
- Church
- Power Cables
- Flooding Zone 2
- School
- River
- Train
- Road
- Surface Water - Flooding
- Abundant Green Space
- Green Space Shortage

Principle
- Ecological Sustainability
- Contextual Responsiveness
- Sustainable Community Building
- Balanced Housing and Mixed Affordability

VISION
- Flood adaptability
- Stronger community connection
- Connection with nature
- Inclusive living
- Sustainable design
- Human-centered mobility
- Improved neighborhood quality of life

CONCEPT
- Flood-adaptive design
- Topography-driven planning
- Pedestrian & cycle-oriented mobility
- Reduced car use
- Green corridor integration
- Mixed-use (Commercial + Live-Work)
- Sustainable human–nature integration
- Socially inclusive housing (Affordable + Cohousing + Private)
- Universal accessibility design
- River view optimization
- Green corridor connectivity
- Native landscape preservation
- Existing path reuse



Case Study: **Marmalade Lane**
Location: Cambridge
Type: Co-housing development
Approximately 40+ homes
Architects: Mole Architects
Completed: 2018

Marmalade Lane is based on a co-housing model, which combines:

Private dwellings
Shared communal spaces

1. Strong Communal Spaces
Shared kitchen and dining (common house)
Communal gardens
Play areas

2. Design for Social Interaction
Pedestrian-focused layout
Houses facing shared spaces
Reduced car dominance

3. Mix of Unit Types
Range from 1-bedroom to family houses
Supports a diverse community



Case Study: **LILAC**
Location: Leeds
Type: Co-housing / Community-led housing
20 homes
Architects: White Design
Completed: 2013

LILAC (Low Impact Living Affordable Community) is a co-housing project based on:

Affordable living
Community collaboration
Sustainable design

1. Shared Spaces
Common house (kitchen + dining)
Shared gardens
Food growing areas

2. Affordable Housing Model
Uses a Mutual Home Ownership Society (MHOS) model

3. Community-Oriented Layout
Homes arranged around shared courtyards
Encourages interaction and visibility


* I have decided to transform the existing light industrial building, currently used for manufacturing, storage, logistics, and administrative functions, into a primary school and library, while incorporating selected existing materials and construction systems into the new design proposal.

Material / Building System	Suitable for School?	Suggested Use in School
Steel Structure	✅	Sports hall, library, cafeteria, assembly hall, canopy
Glass Curtain Wall / Large Glazing	✅	Main entrance, reception, library, art room, corridors facing courtyard
Metal Mesh / Perforated Screen	✅	South/west facades, staircases, terraces, semi-open spaces
Concrete	✅	Foundation, stair core, retaining walls, high-traffic corridors, landscape elements
Metal Roof System	✅	Roofs of halls, sports hall, workshop spaces
Asphalt Paving	Limited use	Parking area, drop-off zone, service access
Metal Cladding Panels	Limited use	Sports hall, service block, rooftop plant room
Industrial Security Fencing	❌	-
Aluminium Window Frames	✅	Windows, curtain wall systems
Large Exposed Hardscape Surfaces	❌	-

Since this area is prone to frequent flooding and is located near a river, the most practical and cost-effective solution to provide electricity to the local community is to install a hydro turbine directly in the river. This approach eliminates the need for large, expensive power pylons and allows more space to be used for the benefit of the residents.

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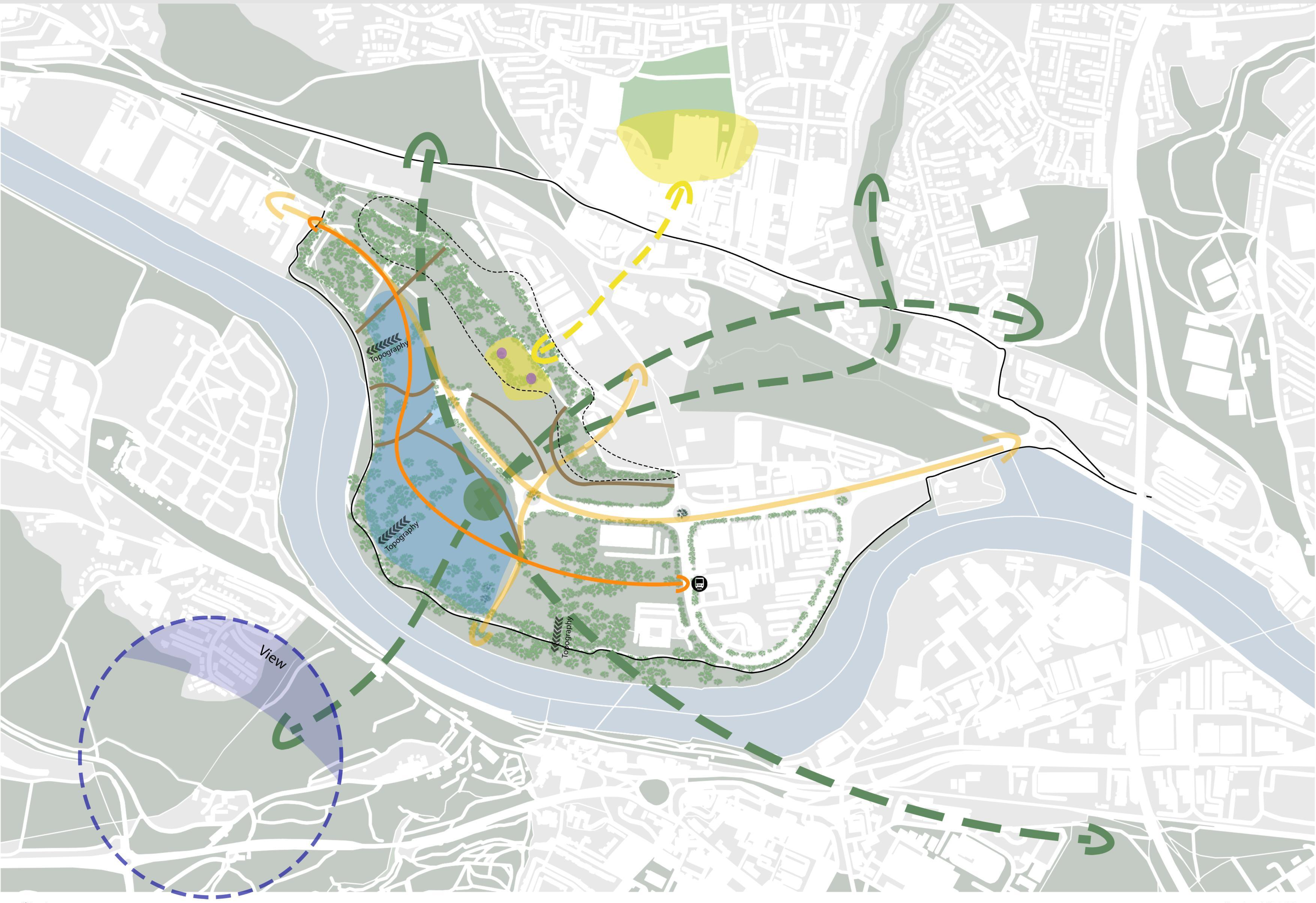
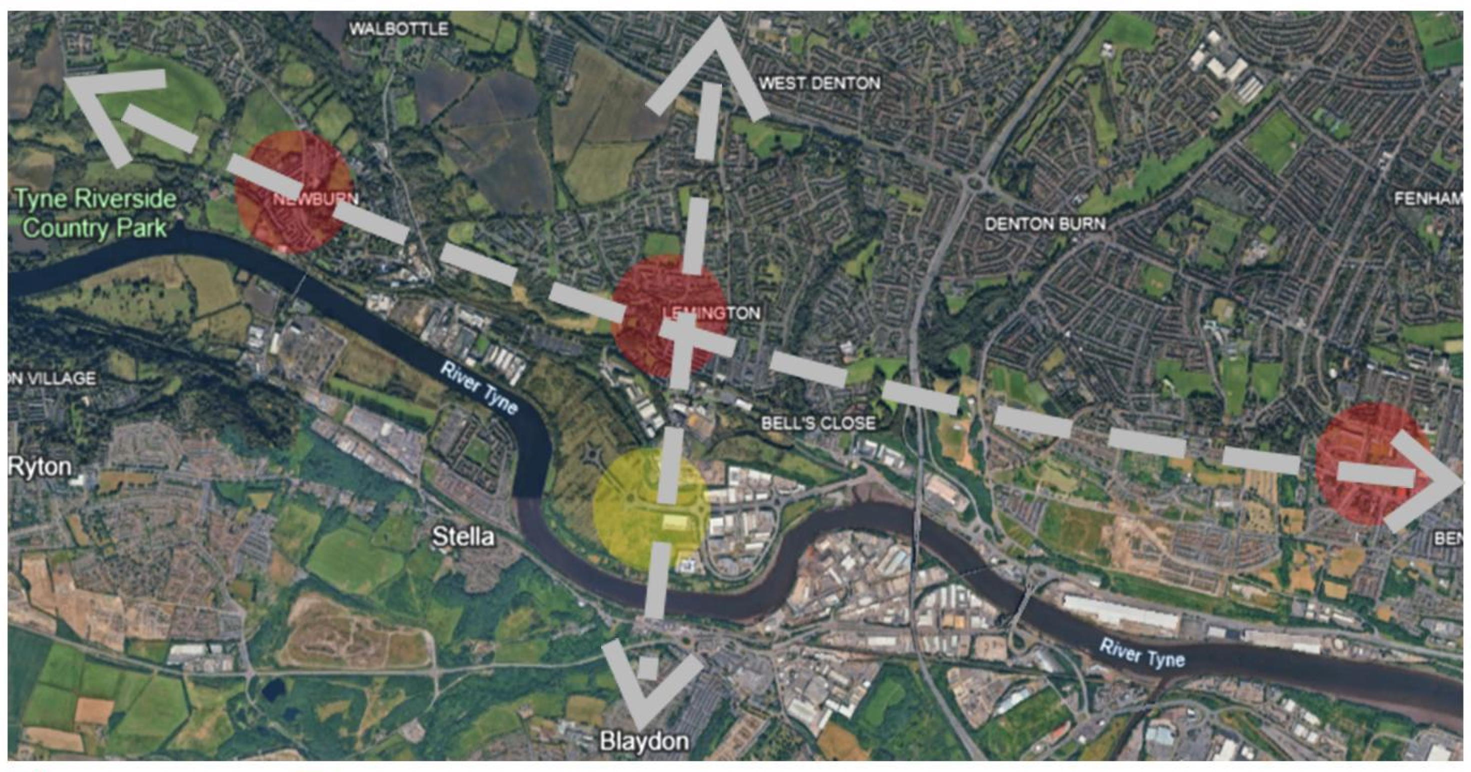


Strategy

Scale: 1/5000

Allotment Connectivity	Main Street	Exicting Path	Bus Station Accessibility	------ Preserve Native Species	Surface	Green Area
Green Corridor	Water Hole	Flow Park	View	—— Cycle Lane	River	



Allotment   Lack of Allotment

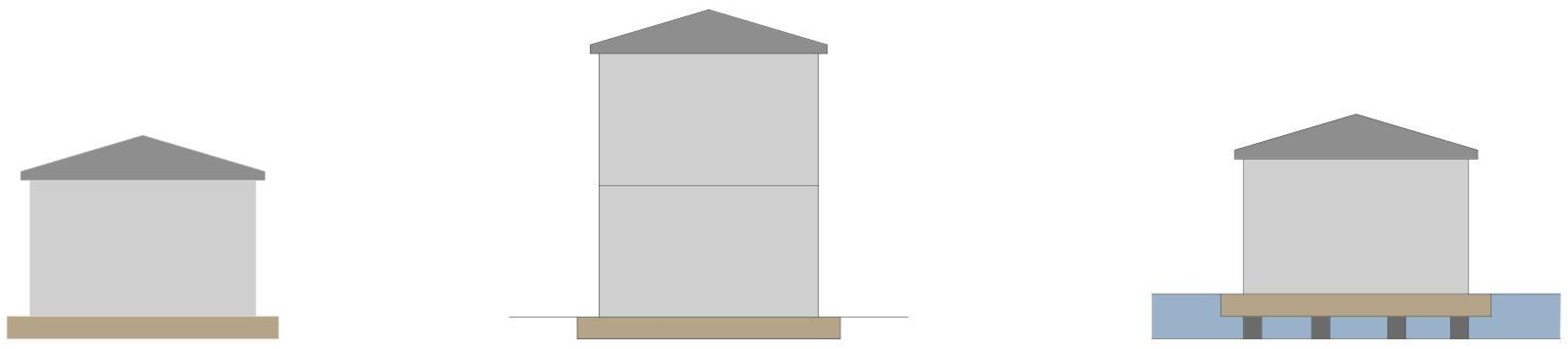

Figure 1. Upton Broad and Marshes landscape. Source: Norfolk Broads Tourism – Upton Broad and Marshes (n.d.)
Figure 2. Water Meadows landscape at Upton. Source: The Watermeadows Project (n.d.)

Water Meadows at Upton is a floodable landscape designed to manage river overflow naturally by allowing excess water to be stored within green open-spaces. During normal conditions the area functions as a public park, while in flood events it protects nearby residential areas by safely absorbing floodwater.


Figure 8. Osney Lock Hydro turbine. Source: CarbonCopy.eco (n.d.)
Figure 9. Osney Lock Hydro in operation. Source: BBC News (2025)

Osney Lock Hydro is a 49kW community-owned hydroelectric project on the River Thames in Oxford. It generates around 188MWh of electricity annually, supplying power to local homes while being managed by a local cooperative.

Figure 3. Amphibious House, designed by Baca Architects. Source: Archello (2015)



The Amphibious House uses a floating foundation supported by buoyant pontoons that allow the building to rise when floodwater enters the dock beneath it. Vertical guideposts fix the house in place, enabling it to move safely up and down with changing water levels without drifting.





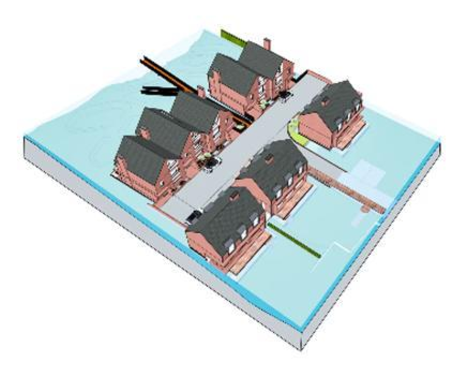


Figure 4,5,6 and 7: Shipston Road Flood-Resilient Homes, UK. Source: Baca Architects

Shipston Road   Location: Stratford-Upon-Avon, UK   Scale: 11 residences, 22,000sqft   Constraints: Flood zones 1, 2 & 3   Architect: BACA

Each home is elevated on piles above potential flood levels, with a controlled, floodable undercroft designed to allow water to pass safely through the site.
A gently ramped access road and raised pedestrian and cycle path ensure safe routes during flood events, while rain gardens and swales absorb surface water and promote biodiversity.

Way

- Car Way
- Cycle Lane
- Pavement
- Green Area
- Water And River

Housing Typology

- 4s1b
- 4s 2b
- 3s Studio
- 2s 2b
- 2s 3b
- 3s 2b
- 3s 3b
- 3s 5b
- 1s 1b
- 1s Studio
- 1s 2b
- 1s 3b
- 2s 4b
- School
- Community Center
- Medical Center
- Shared Area

Land Use and
Spatial Typology

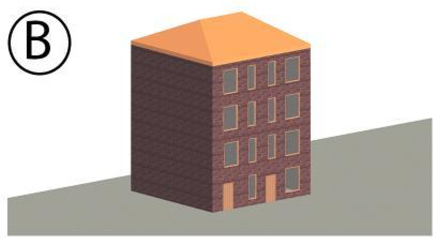
- Apartment
- Live-Work
- House
- School
- Community Center
- Shopping Mall
- Library
- Medical Center
- Semi-Public Open Area
- Pocket
- Allotment

Tenure

- Affordable
- Shared Housing
- Mixed use Housing
- Co-Housing
- Social Housing
- Private Housing
- Accessible Housing

Scale: 1/2500

A


B


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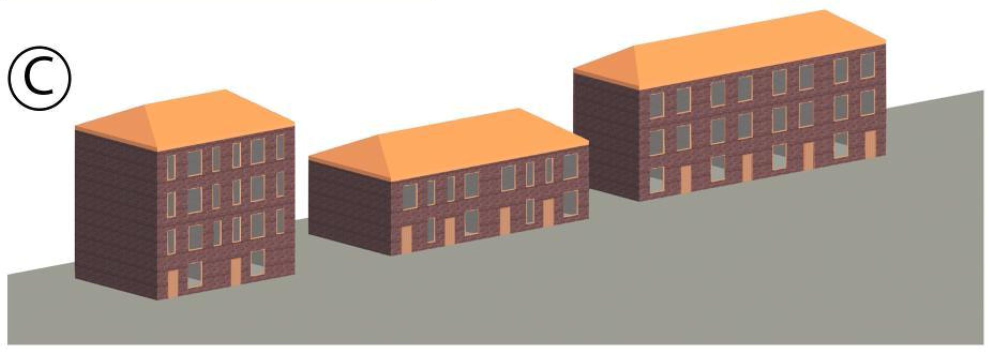

D


E


Housing Mix – Schedule of Accommodation

Housing Type	Units	%
A. Studio	50	5.25%
B. 1B Flat	64	6.72%
C. 2B Flat	323	33.93%
D. 3B Flat	194	20.38%
E. 1B House	50	5.25%
F. 2B House	141	14.81%
G. 3B House	105	11.03%
H. 4B House	17	1.79%
I. 5B House	8	0.84%
Total	**952**	**100%**

Tenure Mix

Tenure Type	Units	%
Private	147	15.4%
Affordable	239	25.1%
Co-Housing	246	25.8%
Social Housing	244	25.6%
Shared Housing	24	2.5%
Mixed Use Housing	52	5.5%
Total	**952**	**100%**

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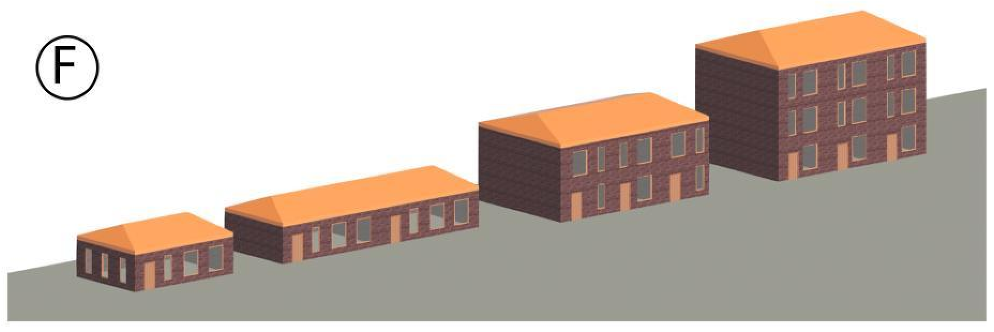

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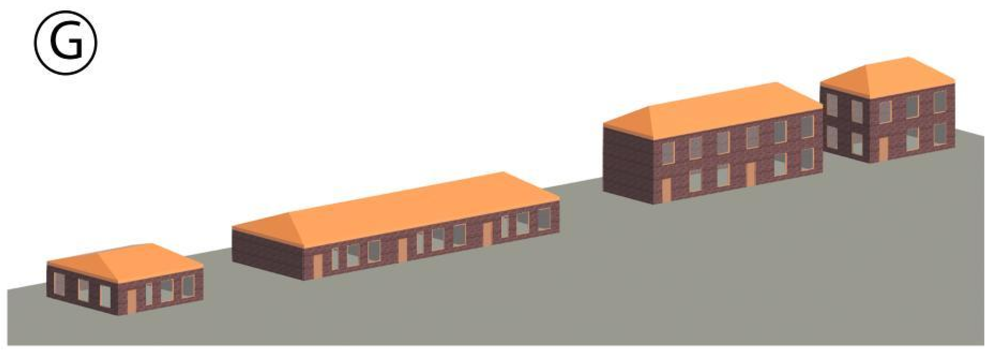

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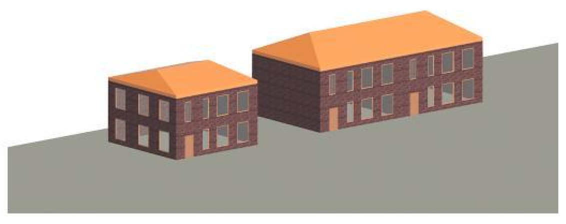

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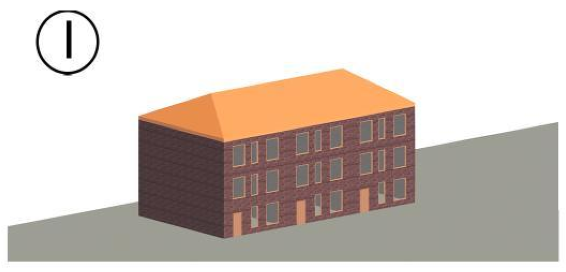
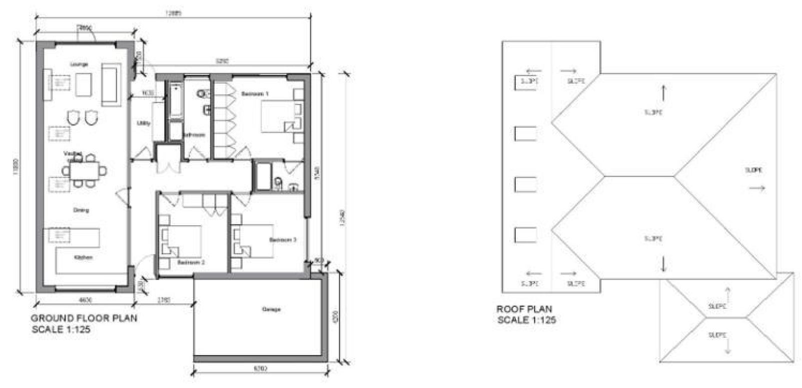
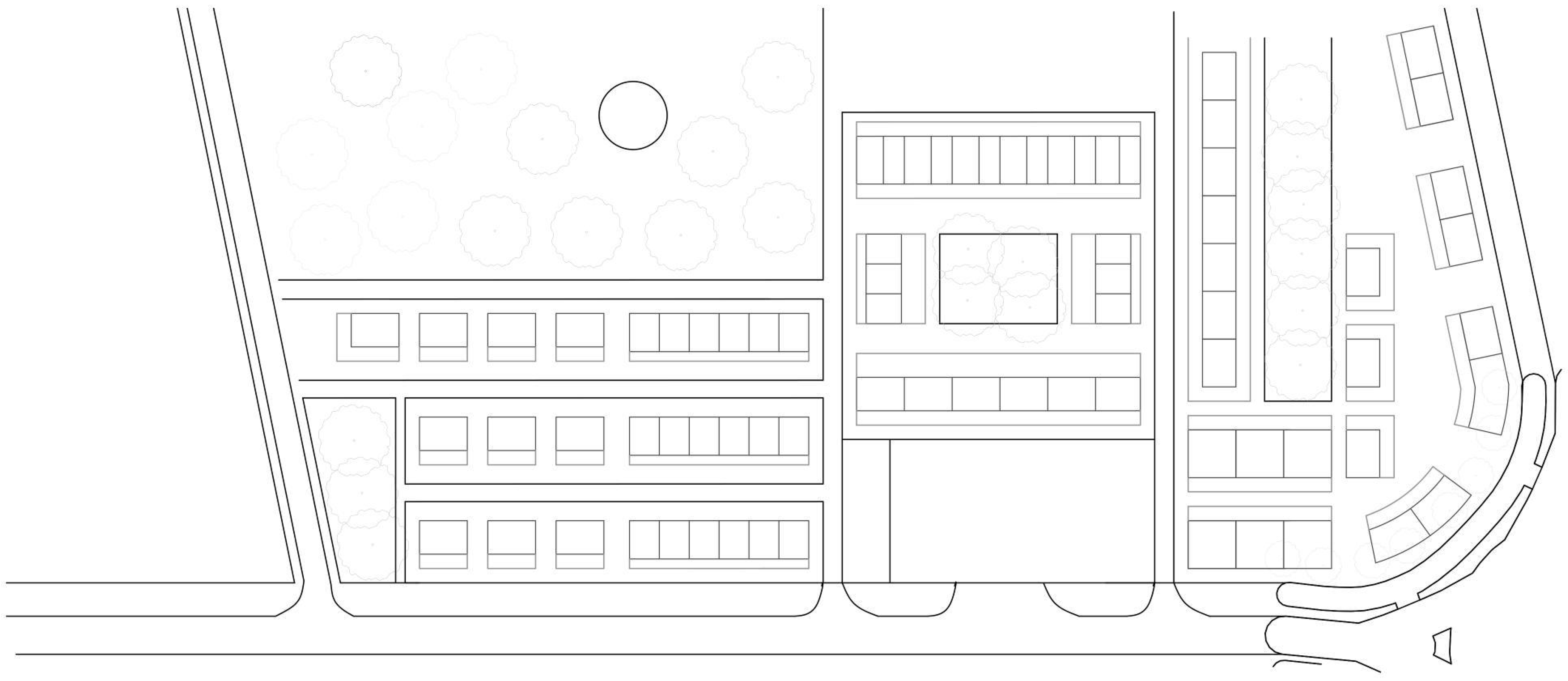


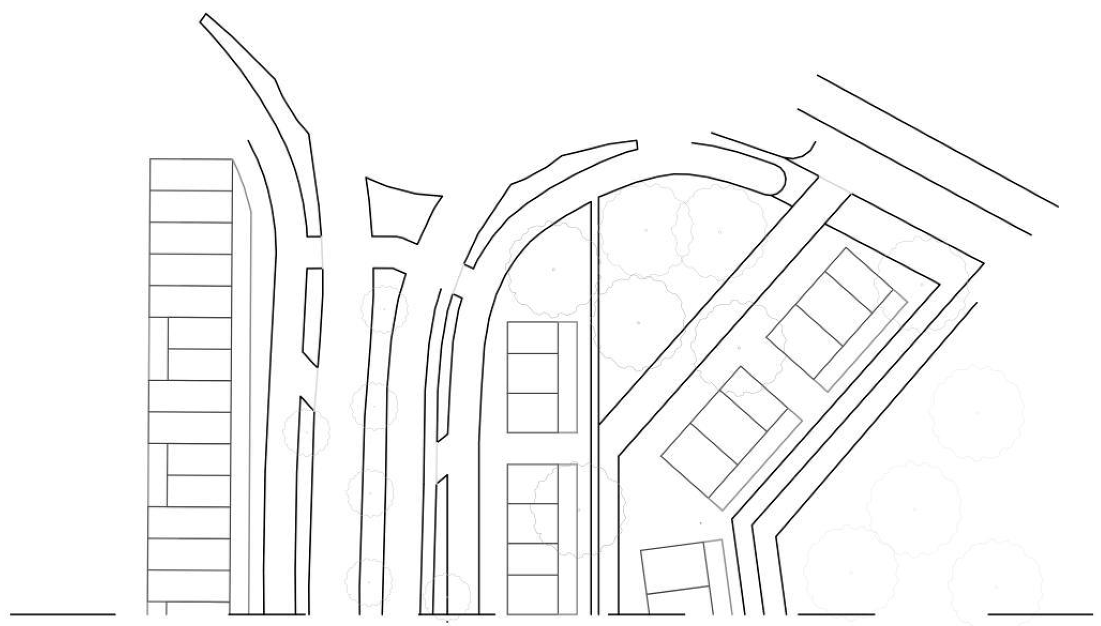
Architect: Alma-nac
Location: London
Type: Narrow townhouse (infill house)
Number of storeys: 3-storey
Bedrooms: 2-bedroom

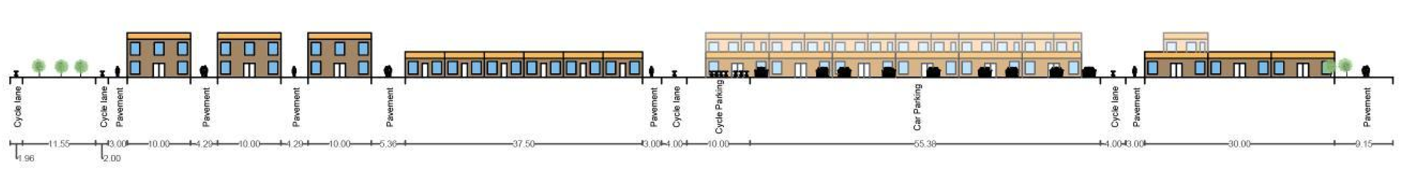
Architect: Fusion 13
Location: York
Type: Self-build-5-bedroom-house
Number of storeys: 3-storey
Bedrooms: 5-bedroom

Architect: Solo Timber Frame Ltd
Location: Bungalow
Type: NEW DESIGN – Reservoir 3 Bedroom
Number of storeys: 1-storey
Bedrooms: 3-bedroom









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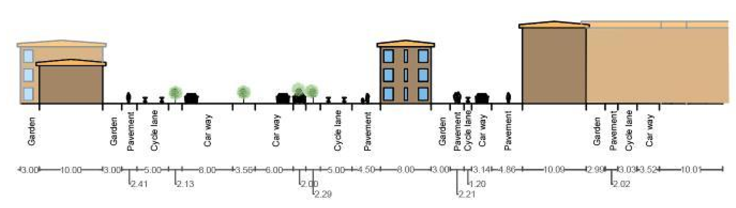

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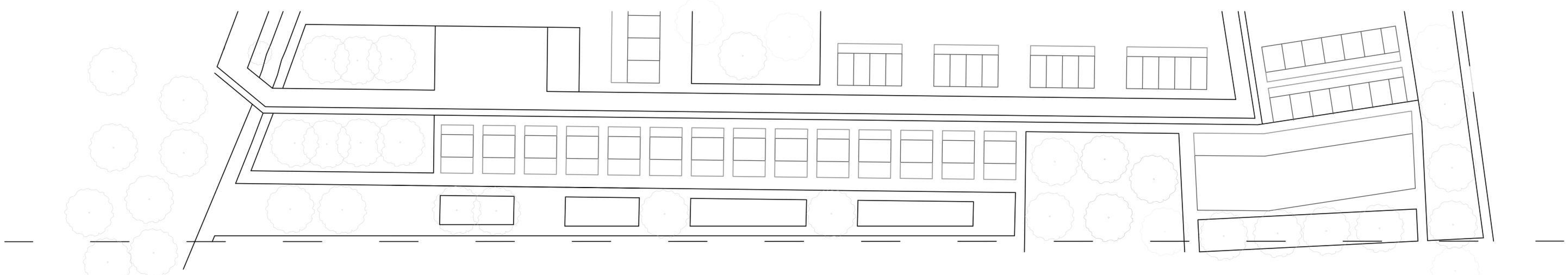

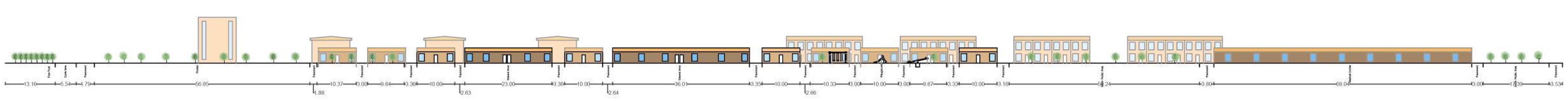

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